

Christopher Kearl

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EDUCATION

Master of Science, Information Systems Management

Aug 2023 – Apr 2025

Brigham Young University - Marriott School of Business

Provo, UT

- GPA: 3.89 / 4.00 (Program), 3.67 / 4.00 (Overall)
- Co-President, Association for Information Systems | Research Assistant, AI Modeling and API infrastructure

Specialized Coursework: Enterprise Programming (JS/TS, React, React Native, Swift, Python, C#), Machine Learning, Cloud Platform Engineering, Managing Tech Startups, Networking Systems, Offensive/Defensive Cybersecurity

EXPERIENCE

Compliance Automation Software Engineer

Jun 2024 - Present

Marriott International

Bethesda, MD

Technologies Used: Python, FastAPI, TypeScript, React, NextJS, TailwindCSS, AWS Suite, Docker, Harness, Terraform

- Engineer organization-wide Compliance-ARC API as sole software engineer, providing security/compliance data, cloud service metadata, and vulnerability tracking across multi-cloud infrastructure to 30+ internal development teams
- Develop Python automation scripts for security remediation at scale, including the patching of 2,300+ EC2 instances with conditional security group updates based on tags, IP ranges, and ownership criteria
- Built React dashboard cataloging deployment requirements across 4 cloud platforms, standardizing compliance docs
- Restored 20 properties and 39 servers during CrowdStrike outage, enabling processing for 35,000+ guests

Adjunct Professor (IS 120)

Dec 2023 – May 2025

Brigham Young University

Provo, UT

Technologies Used: CSS, Tailwind, JavaScript, REST API, Git, AWS API Gateway, Github Classroom, HTML5

- Restructured entire curriculum from scratch, creating course materials (lectures, assignments, and exams) aligned with current industry standards including mobile responsivity, ES6 JavaScript, and REST APIs
- Taught 60+ students with live coding demonstrations, achieving 84% satisfaction (5 points above department average)

Data Software Engineer Intern

May 2023 – Aug 2023

Pattern

Lehi, UT

Technologies Used: Ruby, Rails, Typescript, MySQL, Snowflake, React, Airflow, Airbrake, AWS S3, Postman

- Developed dynamic search keyword filters to categorize keywords by tracking period, resulting in an aggregate average of \$0.50 in savings per advertisement space bid for partners while improving internal ROI analysis
- Reduced Amazon ad metrics dashboard load times by ~3 seconds by optimizing SQL generation and access controls

PROJECTS

KeyCoach

github.com/keycoach/keycoach

Stack: TypeScript, React, Next.js, ML5.js, TensorFlow.js, Phaser JS, 3D Modeling/Printing

- Built web application using ML5.js hand-tracking model to analyze typing finger positioning in real-time via custom 3D-printed camera mirror attachment, providing corrective feedback across typing tests, lessons, and gamified practice
- Led 8-month capstone project team through design, development, and user testing phases

Lecture Transcriber

github.com/ckearl/lecture-transcriber

Stack: Python, Next.js, TypeScript, OpenAI Whisper API, Google Gemini API, Supabase

- Built full-stack lecture transcription app using OpenAI Whisper and Gemini for AI-powered summaries, featuring synchronized audio playback with auto-scrolling transcription in Next.js

Kearl

github.com/ckearl/kearl

Stack: Go, SQLite, GraphQL, REST APIs, Cobra, Viper, Bubble Tea, Lipgloss, Chroma, gojq

- Developed curl alternative in Go featuring interactive JSON explorer, intelligent response caching with TTL, auto-retry with exponential backoff, and environment-specific profiles with template variables for streamlined API development

SKILLS & ACHIEVEMENTS

- AWS Certified Cloud Practitioner (CCP) - May 2024 | Professional Scrum Master I (PSM I) - Oct 2023
- Proficient in TypeScript (React), Python, TailwindCSS, AWS, Swift, SwiftUI, Scrum, Git, JavaScript, MySQL, Node.js
- Experienced in Rust, Go, R, Ruby (Rails), Figma, MongoDB, Node.js, Django, FastAPI, C#, ASP.NET, C++, Tableau
- Fluent in Korean Language
- Former Guitarist/Pianist of band 'Ritt Momney' with multi-millions of total streams on music platforms